Humidity, Feels-Like Temperature, and Precipitation

1. Introduction to Humidity

Humidity refers to the **amount of water vapor** present in the air. It plays a key role in weather, human comfort, and precipitation formation. The amount of moisture in the air depends on **temperature, location, and atmospheric conditions**.

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2. Types of Humidity

a) Absolute Humidity (AH)

- **Definition:** The total mass of water vapor present in a given volume of air.
- **Formula:** AH=Mass of Water Vapor (g)/Volume of Air (m^3)
- **Example:** If a cubic meter of air contains **15 grams of water vapor**, the absolute humidity is **15 g/m³**.

Higher AH means more moisture in the air, but it doesn't account for temperature.

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b) Specific Humidity (SH)

- **Definition:** The **mass of water vapor per unit mass of air** (including both dry air and water vapor).
- **Formula:**SH=Mass of Water Vapor/Total Mass of
- **Example:** If an air parcel contains **1 gram of water vapor per 1 kg of air**, then the specific humidity is **1 g/kg**.

Specific humidity is important for large-scale atmospheric studies because it remains constant as air rises or sinks.

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c) Relative Humidity (RH)

- **Definition:** The **percentage of water vapor** in the air compared to the maximum amount the air can hold at a given temperature.
- **Formula:**RH=(Actual Water Vapor Content/Maximum Water Vapor Capacity)x100
- **Example:**
    - If air at **30°C** can hold **30 g/m³** of water vapor but contains only **15 g/m³**, the RH is: RH=(15/30)×100=50%

Relative humidity depends on temperature—warm air can hold more moisture, so RH decreases as temperature rises.

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3. Feels-Like Temperature (Heat Index & Wind Chill)

The **feels-like temperature** is how the temperature **actually feels to humans**, considering humidity, wind speed, and actual temperature.

a) Effect of Humidity on Feels-Like Temperature

- High humidity **reduces sweat evaporation**, making it harder for the body to cool down.
- The **Heat Index** combines **temperature and RH** to estimate the perceived temperature.
  - **Example:**
    - 35°C with **10% RH** feels like **33°C** (less humid, evaporative cooling works well).
    - 35°C with **80% RH** feels like **45°C** (humid, body cannot cool efficiently).

b) Effect of Wind on Feels-Like Temperature

- In **cold weather**, **wind increases heat loss** from the skin, making it feel colder.
- **Wind Chill Index** is used to determine the **cooling effect of wind**.
  - **Example:**
    - If it's **0°C with no wind**, it feels like **0°C**.
    - If it's **0°C with 40 km/h wind**, it feels like **-10°C**.

"Feels like" temperature is important for health and safety, as extreme heat or cold can cause dehydration or frostbite.

4. Types of Precipitation

a) Rain

- Liquid water droplets falling when **temperature remains above freezing**.
- **Example:** Common in tropical and temperate regions.

b) Snow

- Ice crystals that form when temperatures are **below 0°C** throughout the atmosphere.
- **Example:** Heavy snowfall in **Canada, Russia, and the Himalayas**.

c) Sleet (Ice Pellets)

- Rain that **freezes into ice pellets** before reaching the ground.
- **Example:** Common in **winter storms in the U.S.**

d) Freezing Rain

- Rain that **freezes upon contact** with cold surfaces, forming **ice layers**.
- **Example:** Causes **icy roads and power outages** in winter.

e) Hail

- Ice balls formed in strong **thunderstorm updrafts**.
- **Example:** Found in severe storms, especially in **Kansas (USA) and Punjab (Pakistan)**.

Precipitation is driven by condensation and cooling of humid air. The type depends on temperature at different altitudes.

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5. Monsoon in South Asia

The **South Asian monsoon** is a **seasonal wind reversal** that brings heavy rains in summer and dry conditions in winter.

a) Summer Monsoon (June–September)

- **Cause:**
  - The **Indian subcontinent heats up**, creating a **low-pressure system**.
  - Moist air from the **Indian Ocean** moves inland, bringing heavy rain.
- **Effects:**
  - **Flooding in India, Pakistan, Bangladesh**.
  - Essential for **agriculture (rice, wheat, sugarcane farming)**.
- **Example:**
  - **Mumbai, India** receives **over 2,000 mm of rain** during monsoon season.
  - **Lahore, Pakistan** experiences intense **monsoon thunderstorms**.

b) Winter Monsoon (October–March)

- **Cause:**
  - The land cools down, creating **high pressure** over the subcontinent.
  - Dry winds from **Central Asia** bring **cool, dry weather**.
- **Effects:**
  - **Drought conditions in some areas**.
  - **Mild winters in India, Pakistan, and Bangladesh**.

Monsoons are crucial for water supply but also bring risks like floods, landslides, and crop failures.